

👉 XAI: Rule Based Approaches

FAMAF - UNC

February 10, 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Project Proposals ---

- Due next Monday (13th Feb) 11.59pm ET
- 2 page proposal (more details on “Course Logistics” document on canvas) + References
- Today by 5pm ET, we will post:
 - ▶ Project topics and some concrete problems
 - ▶ Sample proposals and final reports from past iterations
 - ▶ LaTeX and Word templates which you will use to write the proposal



Office Hours and Paper Presentations ---

- Office hours switch this week
 - ▶ Suraj and Jiaqi today
 - ▶ Hima on Thursday
- Students signed up for presentations next week should see us in office hours this week
 - ▶ Full slide deck (ideally!)
 - ▶ An overview of what you plan to present

🍷 Rule Based Approaches ---

Agenda ---

- Paper 1: Interpretable Rule Lists (Letham et. al.)
- Paper 2: Interpretable Rule Sets (Lakkaraju et. al.)
- Discussion

Paper 1 ---



Contributions ---

- Introducing a generative model called **Bayesian Rule Lists (BRL)**
 - ▶ Goal is to output a decision list (if then else-if)
- Novel prior structure to **encourage sparsity**
- Predictive accuracy **on par** with top algorithms

🍷 Decision List: Example ---



Introduction: BRL ---

- Produces a posterior distribution over permutations of if.. then.. Else-if.. rules from a large set of pre-mined rules
- Decision lists with high posterior probability tend to be both accurate and interpretable
 - ▶ Prior favors concise lists with small number of rules and fewer terms in left hand side

Introduction: BRL ---

- New type of **balance** between accuracy, interpretability, and computation
- **What about using other similar models?**
 - ▶ Decision trees (CART)
 - ▶ They employ greedy construction methods
 - ▶ Not particularly computationally demanding but affects quality of solution – both accuracy and interpretability

Pre-mined Rules ---

- A major source of practical feasibility: pre-mined rules
 - ▶ Reduces model space
 - ▶ Complexity of problem depends on number of pre-mined rules
- As long as pre-mined set is expressive, accurate decision list can be found + smaller model space means better generalization (Vapnik, 1995)

🍷 Pre-mined Rules: Intuition ---



🍷 Preliminaries: Notation ---

- Training data $\{(x_i, y_i)\}_{i=1}^n$, $x_i \in \mathbb{R}^d$, $y_i \in \{1, \dots, L\}$

$$\mathbf{x} = (x_1, \dots, x_n) \quad \mathbf{y} = (y_1, \dots, y_n)$$

- Two labels: stroke or no stroke

🍷 Bayesian Decision Lists ---



🍷 Preliminaries: Multinomial ---

- Sampling from a multinomial:



Preliminaries: Dirichlet ---

- Dirichlet: sampling over a probability simplex
 - ▶ E.g., (0.6, 0.4) is a sample from a Dirichlet distribution;
- K-dimensional Dirichlet has k parameters – any positive number

$$\Theta \sim \text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_m)$$

$$P(\theta_1, \theta_2, \dots, \theta_m) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^m \theta_k^{\alpha_k-1}$$

🍷 Preliminaries: Dirichlet Prior ---

- Conjugate prior for multinomial distribution
- Conjugate prior: posterior in the same family as prior
- Prior:

$$(p_1, \dots, p_k) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

- Posterior:

$$(p_1, \dots, p_k) | (x_1, \dots, x_k) \sim \text{Dirichlet}(\alpha_1 + x_1, \dots, \alpha_k + x_k)$$

Bayesian Association Rules ---

$$a \rightarrow y \sim \text{Multinomial}(\theta). \quad \theta | \alpha \sim \text{Dirichlet}(\alpha).$$

$$\theta | \mathbf{x}, \mathbf{y}, \alpha \sim \text{Dirichlet}(\alpha + N)$$

$$N = (N_{.,1}, \dots, N_{.,L})$$

Generative Model ---



Prior Probabilities ---

$$p(d|\mathcal{A}, \lambda, \eta) = p(m|\mathcal{A}, \lambda) \prod_{j=1}^m p(c_j|c_{<j}, \mathcal{A}, \eta) p(a_j|a_{<j}, c_j, \mathcal{A}).$$

Truncated Poisson:

$$p(m|\mathcal{A}, \lambda) = \frac{(\lambda^m/m!)}{\sum_{j=0}^{|\mathcal{A}|} (\lambda^j/j!)}, \quad m = 0, \dots, |\mathcal{A}|.$$

Ensures that sampled values are within bounds!

Also, ensures expected value is close to λ when there are a large number of pre-mined rules

🍷 Prior Probabilities ---

Another Truncated Poisson,

$$p(c_j | c_{<j}, \mathcal{A}, \eta) = \frac{(\eta^{c_j} / c_j!)}{\sum_{k \in R_{j-1}(c_{<j}, \mathcal{A})} (\eta^k / k!)}, \quad c_j \in R_{j-1}(c_{<j}, \mathcal{A}).$$

$p(a_j | a_{<j}, c_j, \mathcal{A})$ is sampled uniformly from available antecedents with appropriate cardinality.

Likelihood ---

- Likelihood is the product of multinomial probability mass functions for the observed label counts at each rule

$$p(\mathbf{y}|\mathbf{x}, d, \boldsymbol{\theta}) = \prod_{j: \sum_l N_{j,l} > 0} \text{Multinomial}(\mathbf{N}_j | \theta_j),$$

$$\theta_j \sim \text{Dirichlet}(\boldsymbol{\alpha}).$$

Marginalize over θ_j , integrate out the intermediate parameter θ_j

Markov Chain Monte Carlo ---

- Generate a chain of random samples until convergence
- Each random sample is a stepping stone for the next one (chain)
- New samples do not depend on any samples before the previous one (Markov)

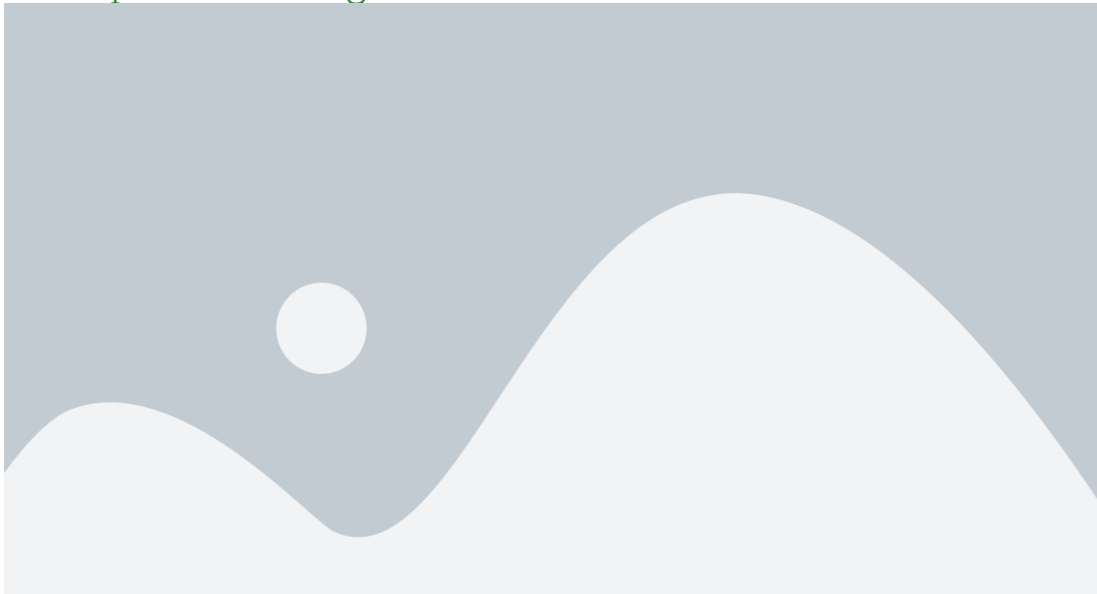
Markov Chain Monte Carlo ---

- How to go to (optimal) d^* from current d^t
- Move an antecedent to a different position in the list
- Add an antecedent that is not currently in the list
- Remove an antecedent from the list

Metropolis Hastings ---

- Start with a random decision list
- Choose a move based on “proposal distribution” Q
- After you choose your move, you compute an acceptance probability A
- Generate a random number u
- If $u \leq A$, then accept; otherwise reject

🍷 Metropolis Hastings ---



🍷 Proposal Probabilities ---

- Move chosen uniformly
- Which antecedents and their new position is also chosen uniformly



🍷 Estimating label of a new observation ---

$$p(\tilde{y} = l | \tilde{x}, d, \mathbf{x}, \mathbf{y}, \alpha) = \frac{\alpha_l + N_{j(d, \tilde{x}), l}}{\sum_{k=1}^L (\alpha_k + N_{j(d, \tilde{x}), k})}.$$

Match the antecedent by looking at feature values of new observation

Experiments ---



Tic-Tac-Toe ---



Stroke Prediction ---

- $N = 12,586$, 14% had stroke
- 6000 times larger than data for CHADS2 score
- Pre-mining: support 10% and max cardinality 2
- 5 fold evaluation

🍷 Stroke Prediction ---



🍷 Stroke Prediction ---



🍷 Stroke Prediction - AUC ---



Paper 2 ---



Contributions ---

- A framework called **Interpretable Decision Sets** (IDS) for classification
- **Novel objective function** + proof of **submodularity**
- Optimization procedure with optimality guarantees
- **Detailed metrics for evaluating interpretability** + user studies

Motivation ---

- Traditional classification models optimize for predictive accuracy
- Very little understanding of the model itself and its predictions
- Model being “readable” is not enough
- Humans should be able to reason about predictions and readily explain the functionality of the model

Decision Lists vs Decision Sets ---

Decision Lists

- Ordered sequence of if-then-else rules
- Order matters: first matching rule applies
- Like a chain of if/else-if statements

Decision Sets

- Unordered collection of if-then rules
- Each rule independently assigns a class
- Rules can overlap (multiple rules may fire)

🍷 Decision Sets ---



🍷 Criteria for Interpretability ---

- **Parsimony**: Fewer rules with fewer conditions
 - ▶ Cognitive limits of human understanding
- **Distinctness**: Minimal overlap of rules w.r.t the data points they cover
 - ▶ No redundant and contradicting explanations of data points
- **Class Coverage**: Explain all the classes in the data
 - ▶ Rules explaining minority classes are important

Problem Formulation ---



Desiderata ---

- We need to optimize for the following criteria
 - ▶ Recall
 - ▶ Precision
 - ▶ Distinctness
 - ▶ Parsimony
 - ▶ Class Coverage
- Recall and Precision $\Rightarrow$ Accurate predictions
- Distinctness, Parsimony, and Class Coverage $\Rightarrow$ Interpretability

Objective Function ---

Parsimony

- Fewer rules: $f_1(\mathcal{R}) = |\mathcal{S}| - \text{size}(\mathcal{R})$
- Fewer predicates: $f_2(\mathcal{R}) = L_{\max} \cdot |\mathcal{S}| - \sum_{r \in \mathcal{R}} \text{length}(r)$

Objective Function ---

Distinctness

- Intra-class overlap:

$$f_3(\mathcal{R}) = N \cdot |S|^2 - \sum_{\substack{r_i, r_j \in \mathcal{R} \\ i \leq j \\ c_i = c_j}} \text{overlap}(r_i, r_j)$$

- Inter-class overlap:

$$f_4(\mathcal{R}) = N \cdot |S|^2 - \sum_{\substack{r_i, r_j \in \mathcal{R} \\ i \leq j \\ c_i \neq c_j}} \text{overlap}(r_i, r_j)$$

🍷 Objective Function ---

Class Coverage

$$f_5(\mathcal{R}) = \sum_{c' \in \mathcal{C}} \mathbf{1} (\exists r = (s, c) \in \mathcal{R} \text{ such that } c = c')$$

Check if there exists some rule corresponding to a given class c

🍷 Objective Function ---

Precision

- Minimize “incorrect” covers:

$$f_6(\mathcal{R}) = N \cdot |\mathcal{S}| - \sum_{r \in \mathcal{R}} |\text{incorrect-cover}(r)|$$

Given a rule $r = (s, c)$, the number of data points which satisfy s but do not belong to class c .

Objective Function ---

Recall

- Encourage at least one “correct” cover per data point:

$$f_7(\mathcal{R}) = \sum_{(\mathbf{x}, y) \in \mathcal{D}} \mathbf{1}(|\{r | (\mathbf{x}, y) \in \text{correct-cover}(r)\}| \geq 1)$$

Given a rule $r = (s, c)$, the number of data points which satisfy s and belong to class c .

Objective Function ---

- Complete objective is

$$\operatorname{argmax}_{\mathcal{R} \subseteq \mathcal{S} \times \mathcal{C}} \sum_{i=1}^7 \lambda_i f_i(\mathcal{R})$$

- The intra-class and inter-class overlap terms are non-monotone
- The parsimony, overlap, and precision terms are non-normal
- All the component terms are submodular

Submodularity ---

Diminishing returns characterization

$$F(A \cup d) - F(A) \geq F(B \cup d) - F(B)$$

Gain of adding d to a small set

Gain of adding d to a large set

Key Property

A non-negative linear combination of submodular functions is submodular

🍷 Objective Function ---

- Complete objective is

$$\operatorname{argmax}_{\mathcal{R} \subseteq \mathcal{S} \times \mathcal{C}} \sum_{i=1}^7 \lambda_i f_i(\mathcal{R})$$

Result

The complete objective is non-negative, non-normal, non-monotone, submodular

🍷 Optimizing the Objective ---

- Maximizing a non-monotone submodular function is **NP-hard**
- **Smooth local search** [SLS] algorithm provides a $2/5$ approximation [Feige, Mirrokni, Vondrak FOCS 07; SIAM Comp. J. 11]
 - ▶ Will be at least $2/5$ of the optimal solution

🍷 Submodular Maximization: Local Search ---





Submodular Maximization: Local Search ---



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Submodular Maximization: Local Search ---





Submodular Maximization: Local Search ---





Submodular Maximization: Local Search ---



🍷 Submodular Maximization: Local Search ---



Local Search ---

- $\sim 1/3$ approximation
 - ▶ At least $1/3$ of optimal solution
- we use a slightly different version of this algorithm
 - ▶ Smooth local search
 - ▶ $2/5$ approximation

🍷 Smooth Local Search ---



Evaluation: Datasets ---



🍷 Evaluating Predictive Performance ---



🍷 Evaluating Goodness of Rules ---

- Results on Medical Diagnosis Data



🍷 Ablation Study ---

- Results on Medical Diagnosis Data



Thank You!



References --- |

- Lakkaraju, Himabindu, Stephen H. Bach, and Jure Leskovec. 2016. “Interpretable Decision Sets: A Joint Framework for Description and Prediction.” In *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 1675–84. ACM.
- Letham, Benjamin, Cynthia Rudin, Tyler H. McCormick, and David Madigan. 2015. “Interpretable Classifiers Using Rules and Bayesian Analysis: Building a Better Stroke Prediction Model.” *The Annals of Applied Statistics* 9 (3): 1350–71.